# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET A

Geography Chapters 5, 6, 7: **Minerals and Energy Resources, Manufacturing Industries, Lifelines of National Economy** |
Class 10 Social Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 12 × 1 = 12

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** mica | Chota Nagpur edge |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A | Location factor |
| 3 | **(b)** electricity | Wide range of applications |
| 4 | **(d)** Khetri mines in Rajasthan | India deficient in copper |
| 5 | **(d)** 1904 | Construction industry |
| 6 | **(c)** gur and khandsari | Second in sugar |
| 7 | **(b)** liberalisation | New models |
| 8 | **(b)** cement | Limestone |
| 9 | **(c)** 1853 | 34 km |
| 10 | **(a)** pipeline | Low running cost |
| 11 | **(c)** Prayagraj and Haldia | 1620 km |
| 12 | **(a)** railways | Largest public sector undertaking |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 13 | A homogeneous, naturally occurring substance with a definable internal structure. | 2 |
| 14 | Agro-based industries use agricultural raw materials (cotton, jute, sugar, edible oil); mineral-based industries use minerals and metals (iron and steel, cement, aluminium, petrochemicals). | 1+1 |
| 15 | It is the only self-reliant industry in India that covers the entire chain from raw material to the highest value-added products. | 2 |
| 16 | Exchange of goods among people, states and countries; international trade is between two countries, local trade is carried on in cities, towns and villages. | 1+1 |
| 17 | Rajdhani Channel, Metro Channel, Green Channel, Business Channel, Bulk Mail Channel and Periodical Channel. | 2 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 18 | Used in manufacturing steel and ferro-manganese alloy (about 10 kg per tonne of steel), bleaching powder, insecticides and paints; production 2018–19 — Madhya Pradesh (33 per cent), Maharashtra (27 per cent), Odisha (16 per cent), Karnataka (12 per cent), Andhra Pradesh (10 per cent). | 3 |
| 19 | Covers transistors, TV, telephones, cellular telecom, exchanges, radars, computers; Bengaluru the electronic capital; other centres — Mumbai, Delhi, Hyderabad, Pune, Chennai, Kolkata, Lucknow, Coimbatore; major impact on employment generation; growth in hardware and software key to success. | 3 |
| 20 | Tourism earns foreign exchange; promotes national integration; supports local handicrafts and cultural pursuits; helps international understanding of our culture and heritage; schemes — Swadesh Darshan 2.0, Vibrant Village Programme, PRASHAD, Paryatan Mitra. | 3 |
| 21 | Physiographic, economic and administrative factors; northern plains — level land, high population, rich agriculture favoured railways; obstacles — wide river beds needing bridges; peninsular hills crossed through gaps or tunnels; Himalayas unfavourable due to high relief, sparse population; sandy plains of Rajasthan, swamps of Gujarat, forests of MP, Chhattisgarh, Odisha, Jharkhand difficult; Sahyadri crossed through ghats. | 3 |
| 22 | Peat — decaying plants in swamps, low carbon, high moisture, low heating capacity; lignite — low-grade brown coal, soft, high moisture (Neyveli); bituminous — buried deep, increased temperature, most popular; metallurgical — high-grade bituminous for blast furnaces; anthracite — highest quality hard coal. | 3 |
| | **OR:** Dust and noxious fumes cause pulmonary diseases; risk of collapsing mine roofs, inundation and fires in coal mines; water sources get contaminated; dumping of waste and slurry degrades land and soil; stream and river pollution; stricter safety and environmental laws needed. | 3 |

| | | |
|---|---|---|
| **23** | Agro-industries raised agricultural productivity; industries depend on agriculture for raw materials; they sell irrigation pumps, fertilisers, insecticides, pesticides, PVC pipes, machines and tools to farmers; this made production efficient. | 3 |
| | **OR:** Capital investment — small scale industry defined by maximum investment allowed on assets (at present ₹1 crore); large scale if more. Bulk — heavy industries like iron and steel (heavy raw materials and products); light industries like electrical goods (light raw materials and products). | 1½+1½ |

## SECTION — D (Long Answer) — Answers

**2 × 5 = 10**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **24** | Minimise water use by reusing and recycling; rainwater harvesting; treat hot water and effluents before release — primary (mechanical: screening, grinding, flocculation, sedimentation), secondary (biological), tertiary (biological, chemical, physical with recycling); regulate overdrawing of groundwater; fit smoke stacks with electrostatic precipitators, fabric filters, scrubbers, inertial separators; use oil or gas instead of coal; silencers on generators; redesign machinery; NTPC's example — latest techniques, ash utilisation, green belts, ash pond management, ecological monitoring. | 5 |
| | **OR:** Ancient hand spinning and handloom; setback in colonial period; first successful mill in Mumbai (1854); boost during world wars; concentrated in cotton belt of Maharashtra and Gujarat due to raw cotton, market, transport, ports, labour, moist climate; close links with agriculture; spinning centralised in Maharashtra, Gujarat, Tamil Nadu, weaving decentralised; world-class spinning but low-quality weaving; handloom, powerloom and mills; khadi as cottage industry. | 5 |
| **25** | Second largest network; roads preceded railways; reasons — lower construction cost, traverse dissected and undulating topography, negotiate high gradients and mountains, economical for few persons and small goods over short distances, door-to-door service lowers loading and unloading cost, feeder to railways, airports and seaports; metalled and unmetalled roads; PMGSY; BRO. | 5 |
| | **OR:** Grown remarkably; foreign exchange; national integration; support to local handicrafts and culture; international understanding; employment; types — heritage, eco, adventure, cultural, medical and business tourism; schemes — Swadesh Darshan 2.0, Vibrant Village Programme, PRASHAD, Paryatan Mitra. | 5 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET B

Geography Chapters 5, 6, 7: **Minerals and Energy Resources, Manufacturing Industries, Lifelines of National Economy** |
Class 10 Social Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 12 × 1 = 12

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** sedimentary rocks | Beds or layers |
| 2 | **(a)** common salt, magnesium and bromine | Most minerals too diffused |
| 3 | **(d)** 10 kg | Ferro-manganese alloy |
| 4 | **(a)** bauxite | Residual deposit |
| 5 | **(c)** 1854 | Cotton textiles |
| 6 | **(c)** Chota Nagpur plateau | Raw materials nearby |
| 7 | **(a)** Bengaluru | Electronics industry hub |
| 8 | **(d)** aluminium smelting | Alumina |
| 9 | **(c)** international trade | Exchange of goods |
| 10 | **(a)** Visakhapatnam | Iron ore outlet |
| 11 | **(d)** Kolkata (Shyama Prasad Mookerjee) | Hoogly |
| 12 | **(a)** Both A and R are true and R is the correct explanation of A | Merits of roads |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 13 | Jharia, Raniganj, Bokaro (any two). | 2 |
| 14 | Alluvial deposits of minerals in sands of valley floors and the base of hills, containing minerals not corroded by water — gold, silver, tin, platinum. | 2 |
| 15 | When hot water from factories and thermal plants is drained into rivers and ponds before cooling, harming aquatic life. | 2 |
| 16 | Construction cost lower than railways; can traverse dissected and undulating topography and high gradients; economical for few persons and small amounts of goods over short distances; door-to-door service; feeder to other modes (any two). | 2 |
| 17 | A scheme under which every village is to be linked to a major town by an all-season motorable road. | 2 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 18 | Sedimentary — in beds or layers formed by deposition, accumulation and concentration in horizontal strata (coal, some iron ore); gypsum, potash salt and sodium salt formed by evaporation in arid regions. Residual — decomposition of surface rocks and removal of soluble constituents leaves a residual mass of weathered material containing ores like bauxite. | 1½+1½ |
| 19 | Energy obtained by altering the structure of atoms, releasing heat used to generate electricity; uranium and thorium available in Jharkhand and the Aravalli ranges of Rajasthan; monazite sands of Kerala are rich in thorium. | 1+2 |
| 20 | Air — undesirable gases like $SO_2$ and CO, dust, smoke from chemical and paper factories, brick kilns, refineries, smelting plants; Water — organic and inorganic wastes and effluents from paper, pulp, chemical, textile, dyeing, refineries, tanneries; Land — dumping of glass, chemicals, salts; Noise — machinery, generators, drills. | 1+1+1 |
| 21 | Provides entertainment and creates awareness about national programmes and policies; includes radio (All India Radio), television (Doordarshan), newspapers in about 100 languages, magazines, books and films; India is the largest producer of feature films. | 3 |
| 22 | Modernises agriculture and reduces dependence on agricultural income by providing jobs in secondary and tertiary sectors; industrial development helps eradicate unemployment and poverty and reduces regional disparities; export of manufactured goods brings foreign exchange; transforming raw materials into high-value goods makes countries prosperous. | 1+1+1 |
| | **OR:** It provides large-scale employment to weavers in their homes as a cottage industry; promotes self-reliance and swadeshi; reduces dependence on mill-made foreign cloth; uses traditional skills. | 3 |

| | | |
|---|---|---|
| **23** | An umbrella programme to prepare India for a knowledge-based transformation; focuses on making technology central to enabling change; IT (Indian Talent) + IT (Information Technology) = IT (India Tomorrow). | 3 |
| | **OR:** Fastest, most comfortable and prestigious mode; covers difficult terrains like mountains, deserts, forests and oceans; helps north-eastern states; Pawan Hans Helicopters serves ONGC offshore operations and inaccessible areas; UDAN promotes regional connectivity and affordable flights. | 3 |

## SECTION — D (Long Answer) — Answers

**2 × 5 = 10**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **24** | Everything from a pin to a building, ship, railway lines, roads, vehicles made of minerals; food contains minerals; toothpaste (silica, limestone, fluorite, titanium oxide, mica). Modes — veins and lodes in igneous/metamorphic rocks; beds or layers in sedimentary rocks; residual mass (bauxite); placer deposits (gold, silver, tin, platinum); ocean waters (salt, magnesium, bromine) and manganese nodules on ocean beds. | 2+3 |
| | **OR:** Petroleum next major energy source after coal; found in anticlines and fault traps of tertiary rocks; Mumbai High, Gujarat (Ankeleshwar most important) and Assam (Digboi, Naharkatiya, Moran-Hugrijan — oldest producer); natural gas with petroleum — Mumbai High and allied fields, Cambay basin, Krishna-Godavari basin; HVJ pipeline (1,700 km) by GAIL; pipeline network expanded to 18,500 km; uses as CNG, PNG, fertilizer and power. | 5 |
| **25** | Twelve major and 200 notified non-major ports; major ports handle 95 per cent of foreign trade; coastline 11098.81 km; west and east coast ports with features (Kandla, Mumbai, JNPT, Mormugao, New Mangalore, Kochi, Tuticorin, Chennai, Visakhapatnam, Paradwip, Kolkata, Haldia). | 5 |
| | **OR:** Trade relations with all major blocks; exports — gems and jewellery, chemicals, agriculture and allied products; imports — petroleum, gems and jewellery, chemicals, base metals, electronics, machinery; India a software giant earning foreign exchange through IT exports; 95 per cent of trade volume by sea; balance of trade; tourism as trade. | 5 |

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## ANSWER KEY & MARKING SCHEME SET C

Geography Chapters 5, 6, 7: **Minerals and Energy Resources, Manufacturing Industries, Lifelines of National Economy** | Class 10 Social Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 12 × 1 = 12

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** Neyveli, Tamil Nadu | Electricity generation |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A | Valuable in electrical industry |
| 3 | **(b)** GAIL (India) | 1,700 km |
| 4 | **(c)** Panchpatmali in Koraput | Aluminium |
| 5 | **(b)** joint sector industry | Jointly owned |
| 6 | **(b)** electronic | Exercise question |
| 7 | **(d)** Bangladesh | Largest producer of raw jute |
| 8 | **(c)** consumer industries | Main role |
| 9 | **(c)** favourable balance of trade | Trade |
| 10 | **(d)** Lahul-Spiti valley | Built by BRO |
| 11 | **(d)** 17 | Railway zones |
| 12 | **(c)** regional air connectivity | Ude Desh ka Aam Nagrik |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 13 | Ferrous minerals contain iron (iron ore, manganese, nickel, cobalt); non-ferrous minerals do not contain iron (copper, bauxite, lead, zinc, gold). | 1+1 |
| 14 | Electricity generated from oceanic tides using floodgate dams across inlets; Gulf of Khambhat, Gulf of Kuchchh or Gangetic delta in Sunderbans. | 1+1 |
| 15 | Production of goods in large quantities after processing raw materials into more valuable products. | 2 |
| 16 | Basic chemicals undergo processing to produce other chemicals used for industrial application, agriculture or consumer markets. | 2 |
| 17 | It rules out trans-shipment losses and delays; initial cost high but running costs minimal; can transport liquids, gases and slurry; made far inland refineries possible. | 2 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 18 | They occur in cracks, crevices, faults or joints; smaller occurrences are veins and larger ones lodes; formed when minerals in liquid/molten and gaseous forms are forced upward through cavities towards the surface where they cool and solidify; e.g. tin, copper, zinc, lead. | 1+1+1 |
| 19 | Organic and inorganic wastes and effluents with dyes, detergents, acids, salts, heavy metals like lead and mercury, pesticides, plastics; control — reusing and recycling water, rainwater harvesting, treating effluents in primary, secondary and tertiary stages before release. | 1½+1½ |
| 20 | Proximity to jute-producing areas; inexpensive water transport supported by railways, roadways and waterways; abundant water for processing raw jute; cheap labour from West Bengal, Bihar, Odisha and UP; Kolkata provides banking, insurance and port facilities for export. | 3 |
| 21 | NW-1: Ganga between Prayagraj and Haldia (1620 km); NW-2: Brahmaputra between Sadiya and Dhubri (891 km); NW-3: West-Coast Canal in Kerala (Kottapuram–Kollam, Udyogamandal and Champakkara canals — 205 km); NW-4: Godavari and Krishna stretches with Kakinada–Puducherry canals (1078 km); NW-5: Brahmani, Matai, Mahanadi delta channels and East Coast Canal (588 km). | 3 |
| 22 | Six-lane highways linking Delhi–Kolkata–Chennai–Mumbai–Delhi; North–South corridor (Srinagar–Kanniyakumari) and East–West corridor (Silchar–Porbandar); reduce time and distance between mega cities; implemented by NHAI. | 3 |
| | **OR:** Constructed and maintained by the Border Roads Organisation (1960) in bordering areas; strategic importance in northern and north-eastern border areas; improved accessibility in difficult terrain; economic development of these areas; e.g. Atal Tunnel connects Manali to Lahul-Spiti throughout the year. | 1+1+1 |

| | | |
|---|---|---|
| **23** | Clay-like substance from which alumina and then aluminium is obtained; formed by decomposition of rocks rich in aluminium silicates; deposits in Amarkantak plateau, Maikal hills, Bilaspur-Katni plateau; Odisha largest producer (Panchpatmali, Koraput); aluminium combines strength with lightness, conductivity and malleability. | 3 |
| | **OR:** Use public transport instead of individual vehicles; switch off electricity when not in use; use power-saving devices; use non-conventional sources; 'energy saved is energy produced'. | 3 |

## SECTION — D (Long Answer) — Answers

**2 × 5 = 10**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **24** | Most abundant fossil fuel; Gondwana coal (a little over 200 million years) — metallurgical coal; Damodar valley (West Bengal–Jharkhand) — Jharia, Raniganj, Bokaro; Godavari, Mahanadi, Son and Wardha valleys; Tertiary coal (about 55 million years) — Meghalaya, Assam, Arunachal Pradesh, Nagaland; lignite at Neyveli (TN); coal is bulky, so heavy industries and thermal power stations are located on or near coalfields. | 5 |
| | **OR:** Hydroelectricity — fast-flowing water, renewable; multi-purpose projects like Bhakra Nangal, Damodar Valley Corporation, Kopili Hydel Project. Thermal — burning coal, petroleum and natural gas; non-renewable fossil fuels; per capita electricity consumption is an index of development; nuclear power stations; solar and wind plants. | 5 |
| **25** | Source of raw material — agro-based (cotton, jute, sugar) and mineral-based (iron and steel, cement); main role — basic/key (iron and steel) and consumer (sugar, toothpaste); capital investment — small scale (investment up to ₹1 crore) and large scale; ownership — public (BHEL, SAIL), private (TISCO, Bajaj), joint (OIL), cooperative (sugar in Maharashtra, coir in Kerala); bulk — heavy (iron and steel) and light (electrical goods). | 5 |
| | **OR:** Basic industry — all other industries depend on it for machinery; used for engineering goods, construction, defence, medical, telephonic, scientific equipment; production and consumption of steel is an index of development; heavy industry — raw materials and finished goods bulky; iron ore, coking coal, limestone 4:2:1, manganese; ideally located near raw materials with efficient transport; Chota Nagpur — low-cost iron ore, high-grade raw materials in proximity, cheap labour, vast home market. | 5 |

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